Report on

DEEPFAKE DETECTION USING HAND-CRAFTED FEATURES AND DEEP LEARNING

Submitted by
Soumadip Singha Mahapatra
(Roll No- 2022ITB097)
B.Tech, Information Technology
Indian Institute of Engineering Science and Technology, Shibpur

Under the guidance of
Dr. Ruchira Naskar
Associate Professor, Dept. of Information Technology
Indian Institute of Engineering Science and Technology, Shibpur

In partial fulfilment of
The requirements of the
Summer Research Internship

(May-July 2025)

Submission Date: October 2025

Acknowledgement

We express our deepest gratitude to **Dr. Ruchira Naskar**, *Associate Professor, Dept. of Information Technology, Indian Institute of Engineering Science and Technology, Shibpur(IIEST)*, for providing us the invaluable opportunity to undertake this **Summer Research Internship** on *"Deepfake Detection Using Hand-Crafted Features and Deep Learning"*. Her guidance, encouragement, and support have been instrumental throughout the course of this work.

We extend our sincere thanks to our **research coordinators** for their consistent support, constructive suggestions, and valuable insights that helped refine our ideas and strengthen the quality of our project.

We would also like to thank the **Department of Information Technology, Indian Institute of Engineering Science and Technology, Shibpur (IIEST)**, for providing us with the academic resources, research environment, and technical exposure essential for carrying out this internship successfully.

Finally, we wholeheartedly acknowledge the contribution of my teammate, **Ankesh Hatui**, whose cooperation, hard work, and team spirit played a significant role in the successful completion of this project.

Soumadip Singha Mahapatra

Index

1. Introduction

1.1 Background and Motivation

Deepfake technology has advanced rapidly, enabling the creation of highly realistic synthetic images and videos using deep learning methods, particularly GANs. While powerful, this technology poses serious risks due to its potential misuse. Deepfakes can manipulate political narratives, enable identity theft, support financial fraud, fabricate legal evidence, and undermine trust in digital media. The increasing realism of such content makes automated detection crucial.

1.2 Project Objectives

This internship project focuses on building a complete deepfake detection system with the following goals:

1. Process multiple real and fake video datasets
2. Detect faces and extract frames efficiently
3. Apply forensic feature extraction methods
4. Develop a multi-modal deep learning detection model
5. Achieve high accuracy with balanced precision–recall
6. Ensure the system is scalable and reproducible
7. Validate performance using cross-validation

1.3 Scope of Work

The project covers the full deepfake detection pipeline:

Data Preparation:

- Load and process videos from multiple sources
- Extract frames and organize them into real/fake classes

Face Extraction:

- Use MediaPipe for frame-wise face detection and cropping
- Maintain mapping between original frames and extracted faces

Feature Engineering:

- Error Level Analysis (ELA) for compression artifacts
- Steganalysis Rich Model (SRM) for frequency patterns
- Discrete Cosine Transform (DCT) coefficient extraction

Deep Learning:

- A dual-stream CNN for multi-modal feature learning
- Single-stream CNN variants (ELA, SRM, DCT) for comparison
- Memory-efficient generators and K-fold cross-validation

1.4 Report Organization

Section 2 discusses related work; Section 3 covers datasets and methodology. Section 4 explains the processing pipeline, while Sections 5 and 6 detail feature extraction and model architecture. Section 7 describes the experimental setup, Section 8 presents results, Section 9 outlines challenges, and Section 10 concludes with future scope.

2. Literature Review and Background

2.1 Deepfake Generation Techniques

Deepfakes are mainly created using the following deep learning methods:

- **Generative Adversarial Networks (GANs):**
  GANs use a generator–discriminator setup where the generator produces synthetic faces and the discriminator identifies fakes. Through adversarial training, GANs generate highly realistic images.
- **Autoencoders:**
  Autoencoders learn compressed facial representations and reconstruct them, enabling face-swapping by replacing identity features in the decoded output.
- **Face Reenactment:**
  Methods like Face2Face and Neural Textures transfer expressions and head movements from a source video to a target face while preserving the target's identity.

2.2 Deepfake Detection Approaches

Common detection techniques include:

- **Temporal Analysis:**
  Detects unnatural blinking, head movements, or flickering across video frames.
- **Physiological Signals:**
  Uses subtle skin color changes (e.g., heart-rate-based signals) to identify inconsistencies not reproduced in deepfakes.
- **Frequency Domain Analysis:**
  Examines frequency patterns that reveal GAN-related artifacts invisible in raw images.
- **Deep Learning Methods:**
  CNNs such as XceptionNet, EfficientNet, or custom multi-stream models learn discriminative features to classify real vs. fake faces.

2.3 Hand-Crafted Forensic Features

- **Error Level Analysis (ELA):**
  Highlights differences in compression errors to expose manipulated or recompressed regions.
- **Steganalysis Rich Model (SRM):**
  Uses high-pass filters to capture noise residuals and statistical patterns, making it effective for detecting subtle manipulations.
- **Discrete Cosine Transform (DCT):**
  Analyzes JPEG frequency coefficients; inconsistencies in DCT values often indicate editing or GAN-generated content.

2.4 Multi-Modal Learning

Combining multiple features improves robustness.
Early, late, or hybrid fusion methods integrate complementary cues, reducing false positives and improving generalization.

This project adopts an early-fusion dual-stream model using ELA and SRM features.

3. Datasets and Methodology

3.1 Dataset Overview

This project uses multiple deepfake video datasets consolidated into several processing batches.

Dataset Sources Used:
- VidTIMIT & DFTIMIT
- Celeb-DF
- FaceForensics++ (DF)
- FaceForensics++ (FS)
- FaceForensics++ (F2F)
- FaceForensics++ (NT)
- DFDC (DeepFake Detection Challenge Dataset)

3.2 Overall Methodology

The project follows a structured end-to-end deepfake detection pipeline.

Phase 1: Data Acquisition and Organization
- Collect datasets from all listed sources
- Arrange folders systematically

Phase 2: Video Processing
- Extract frames from each video at fixed intervals
- Detect faces using MediaPipe
- Crop and save detected face regions
- Maintain a mapping between frames and cropped faces

Phase 3: Feature Extraction
- Apply ELA to generate ELA feature images
- Apply SRM filters to generate noise residual maps
- Perform DCT analysis to extract frequency coefficient matrices
- Preserve structured directory hierarchy for all feature types

Phase 4: Data Preparation
• Create paired input paths for ELA and SRM images
• Use an 80/10/10 train, validation, and test split
• Obfuscate filenames to prevent metadata leakage
• Use custom memory-efficient data generators

Phase 5: Model Development and Training
• Build a dual-stream CNN for multimodal learning
• Train using 10-fold cross-validation
• Use dropout, batch normalization, and learning rate control
• Track accuracy, loss, and other training metrics

Phase 6: Evaluation and Analysis
• Compute accuracy, precision, recall, and F1-score
• Generate confusion matrices
• Check fold-wise performance consistency
• Visualize training curves and final evaluation metrics

3.3 Workflow Summary

Videos → Frame Extraction → Face Detection → ELA, SRM, DCT Feature Processing → Data Pairing → Train/Validation/Test Split → Path Obfuscation → Dual-Stream CNN → Training → Evaluation → Results

4. Hand-Crafted Feature Extraction

4.1 Error Level Analysis (ELA)

ELA is a forensic technique used to detect image manipulation by analyzing JPEG compression artifacts. When an image is manipulated, different regions are compressed unevenly, creating detectable inconsistencies.

How ELA Works:

1. The image is re-saved at a known JPEG quality level.
2. The difference between the original and recompressed image is calculated.
3. Regions with higher error values appear brighter, indicating possible manipulation.
4. The difference image is enhanced to make artifacts more visible.

What ELA Detects:

- Splicing and copy-move forgeries
- Local edits and retouching
- Compression inconsistencies introduced by deepfake generation
- Artifacts from recompressed video frames

ELA is applied to entire directories of images while preserving the folder structure.

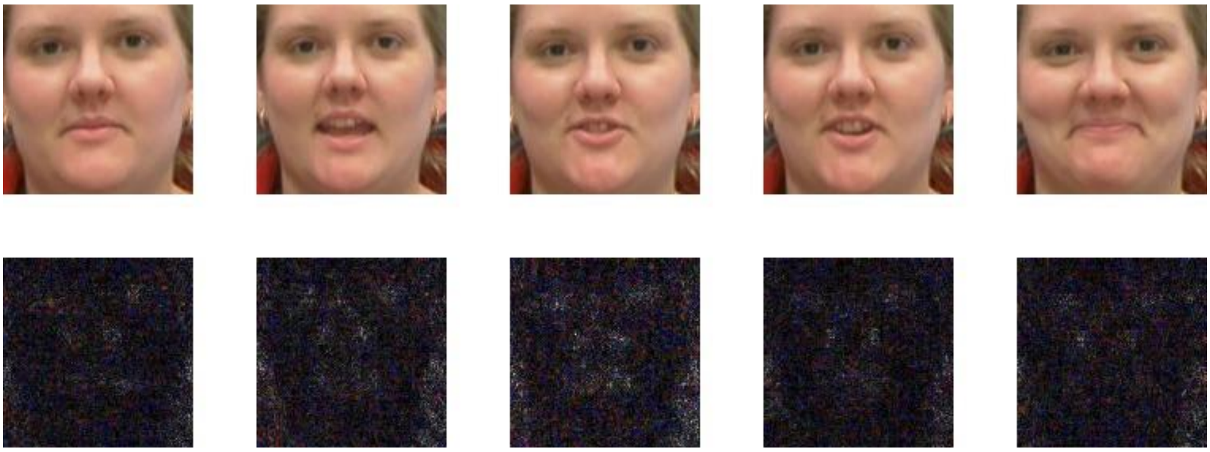


Fig: Authentic Image and Its ELA

4.2 Noise Residuals from Spatial Rich Model (SRM)

SRM extracts **noise residuals** from images using a fixed bank of **30 high-pass filters**. These filters highlight statistical irregularities that occur when an image is manipulated or GAN-generated.

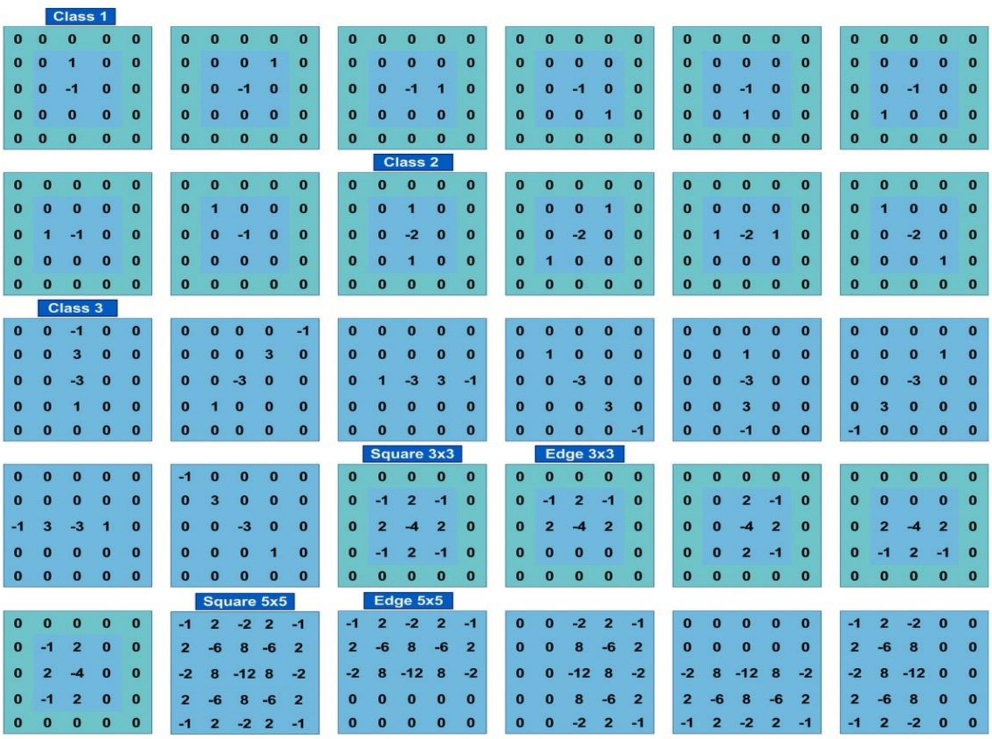
SRM Process:

1. Each SRM filter is convolved with the image to extract noise patterns.
2. Responses from all filters are combined into a residual map.
3. The output is normalized and saved as an SRM feature image.

What SRM Captures:

- High-frequency artifacts
- GAN fingerprints
- Resampling and interpolation traces
- Noise inconsistencies caused by editing

SRM Filter Set:



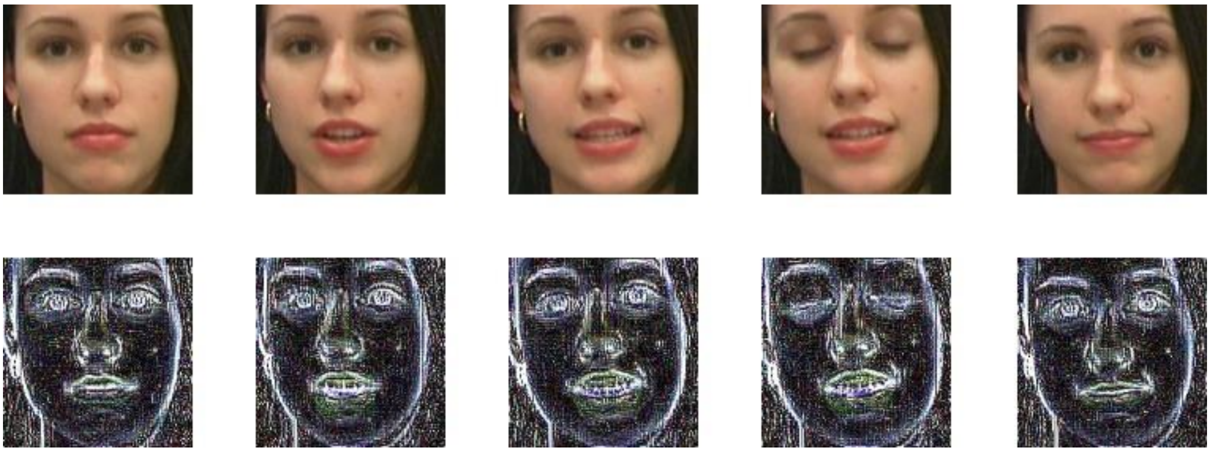


Fig: Authentic Image and Its SRM

4.3 Discrete Cosine Transform (DCT)

DCT examines the frequency components of an image, making it effective for detecting deepfake artifacts.

How DCT Works:

5. The image is divided into frequency components using the DCT.
5. Low-frequency coefficients represent general structure.
5. High-frequency coefficients represent details and noise.
5. Deepfakes often show irregular frequency distributions, especially in mid-frequency ranges.
5. The most informative 16×16 low-frequency block is extracted as the feature representation.

What DCT Reveals:

- GAN-generated frequency anomalies
- Recompression inconsistencies
- Abnormal structural patterns in synthetic faces

Fig: Authentic Image and Its DCT

5. Deep Learning Architecture

5.1 Architecture Overview and Experimental Design

To evaluate the contribution of different forensic features, several CNN variants were developed:

• Single-Stream (ELA only)
• Single-Stream (SRM only)
• Single-Stream (DCT only)
• Dual-Stream (ELA + SRM)
• Dual-Stream (ELA + DCT)
• Dual-Stream (SRM + DCT)

5.2 Single-Stream CNN (Summary with Light Layer Discussion)

Single-stream models take **one forensic feature** as input (ELA, SRM, or DCT).

Input Layer:

• Input Shape: (128, 128, 3) for ELA/SRM or (128, 128, 1) for DCT
• Single input corresponding to one forensic feature

Convolution Layers:

Layer 1: Conv2D
- 32 filters, 3×3 kernel, ReLU, same padding
- Extracts basic low-level patterns

Layer 2: BatchNormalization
- Stabilizes activations

Layer 3: Conv2D
- 32 filters, 3×3 kernel, ReLU, same padding
- Learns more complex feature combinations

Layer 4: BatchNormalization

Layer 5: MaxPooling2D
- Pool size: 2×2
- Downsamples to (64, 64, 32)

Layer 6: Dropout (0.2)
- Reduces overfitting

Flattening and Classification:

Layer 7: Flatten
- Converts feature maps to 131,072-dimensional vector

Layer 8: Dense
- 256 units, ReLU activation

Layer 9: BatchNormalization

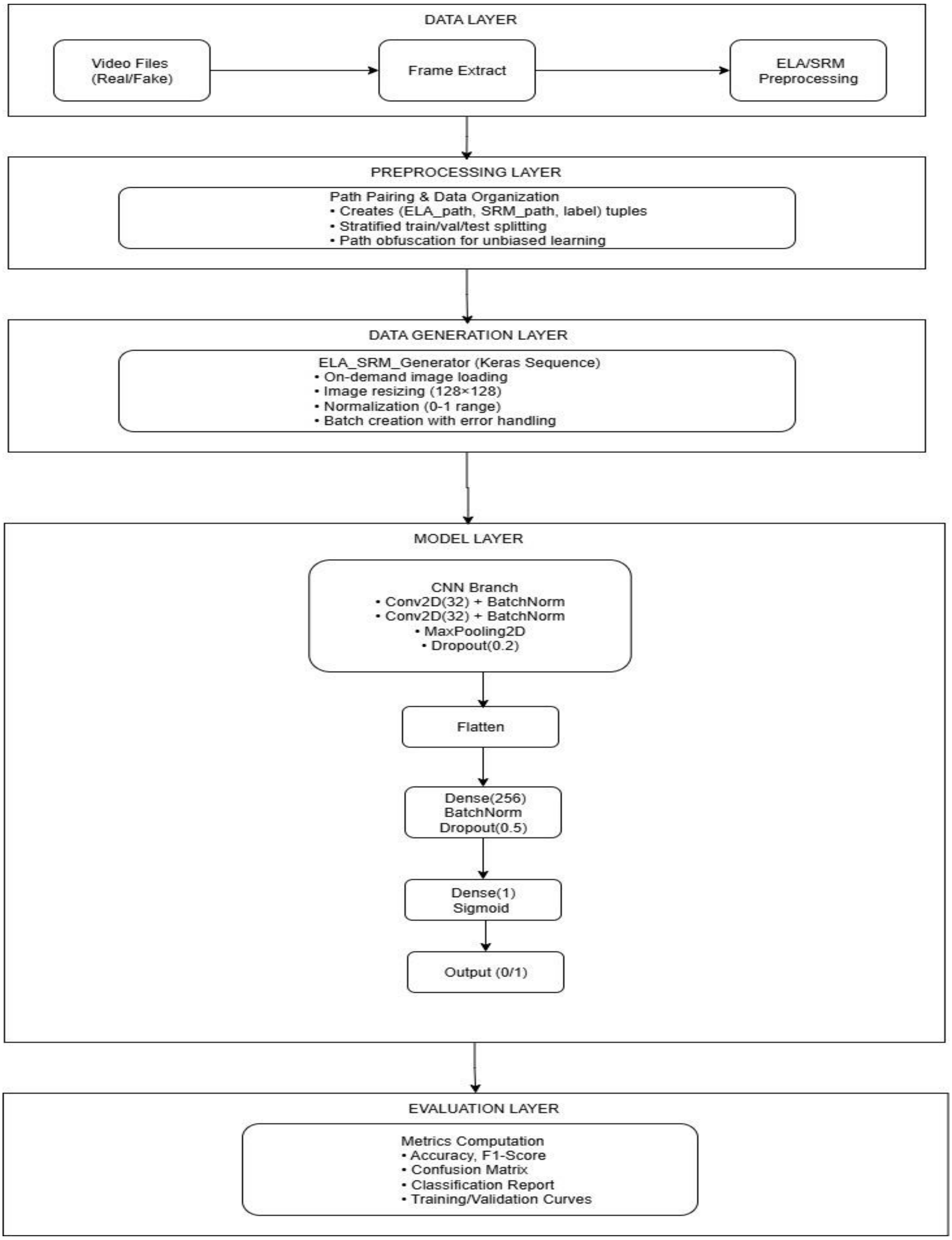
Layer 10: Dropout (0.5)

Layer 11: Dense (Output)

- 1 unit, Sigmoid activation
- Final binary classification (real/fake)

Single-Stream Characteristics:

1. Simpler architecture (one input branch)
2. Fewer parameters (~33M vs ~67M for dual-stream)
3. Faster training and inference (~2× faster than dual-stream)
4. Faster preprocessing (only one feature type needed)
5. Specialized for single forensic technique
6. Baseline for comparison with dual-stream
7. Lower memory requirements (single input buffer)



5.3 Dual-Stream CNN Design

The dual-stream architecture processes ELA and SRM features in parallel. After establishing the single-stream baselines, this model was introduced to evaluate the advantage of combining complementary forensic representations.

Input layer

- Two separate inputs for parallel processing
    - ELA input: (128, 128, 3)
    - SRM input: (128, 128, 3)

ELA branch

Layer 1: Conv2D
- 32 filters, 3×3 kernel, ReLU, same padding
- Extracts low-level ELA features

Layer 2: BatchNormalization
- Stabilizes activations

Layer 3: Conv2D
- 32 filters, 3×3 kernel, ReLU, same padding
- Learns more complex feature patterns

Layer 4: BatchNormalization

Layer 5: MaxPooling2D
- Pool size: 2×2
- Downsamples to (64, 64, 32)

Layer 6: Dropout (0.2)
- Reduces overfitting

SRM branch

Layer 1: Conv2D
- 32 filters, 3×3 kernel, ReLU, same padding
- Extracts low-level SRM features

Layer 2: BatchNormalization
- Stabilizes activations

Layer 3: Conv2D
- 32 filters, 3×3 kernel, ReLU, same padding
- Learns SRM-specific patterns

Layer 4: BatchNormalization

Layer 5: MaxPooling2D
- Pool size: 2×2
- Downsamples to (64, 64, 32)

Layer 6: Dropout (0.2)
- Reduces overfitting

Fusion layer

Layer 7: Concatenate
- Combines features from both branches
- Output shape: (64, 64, 64)

Flattening

Layer 8: Flatten

• Converts fused feature map to 262,144-unit vector

Classification head

Layer 9: Dense

• 256 units, ReLU activation

Layer 10: BatchNormalization

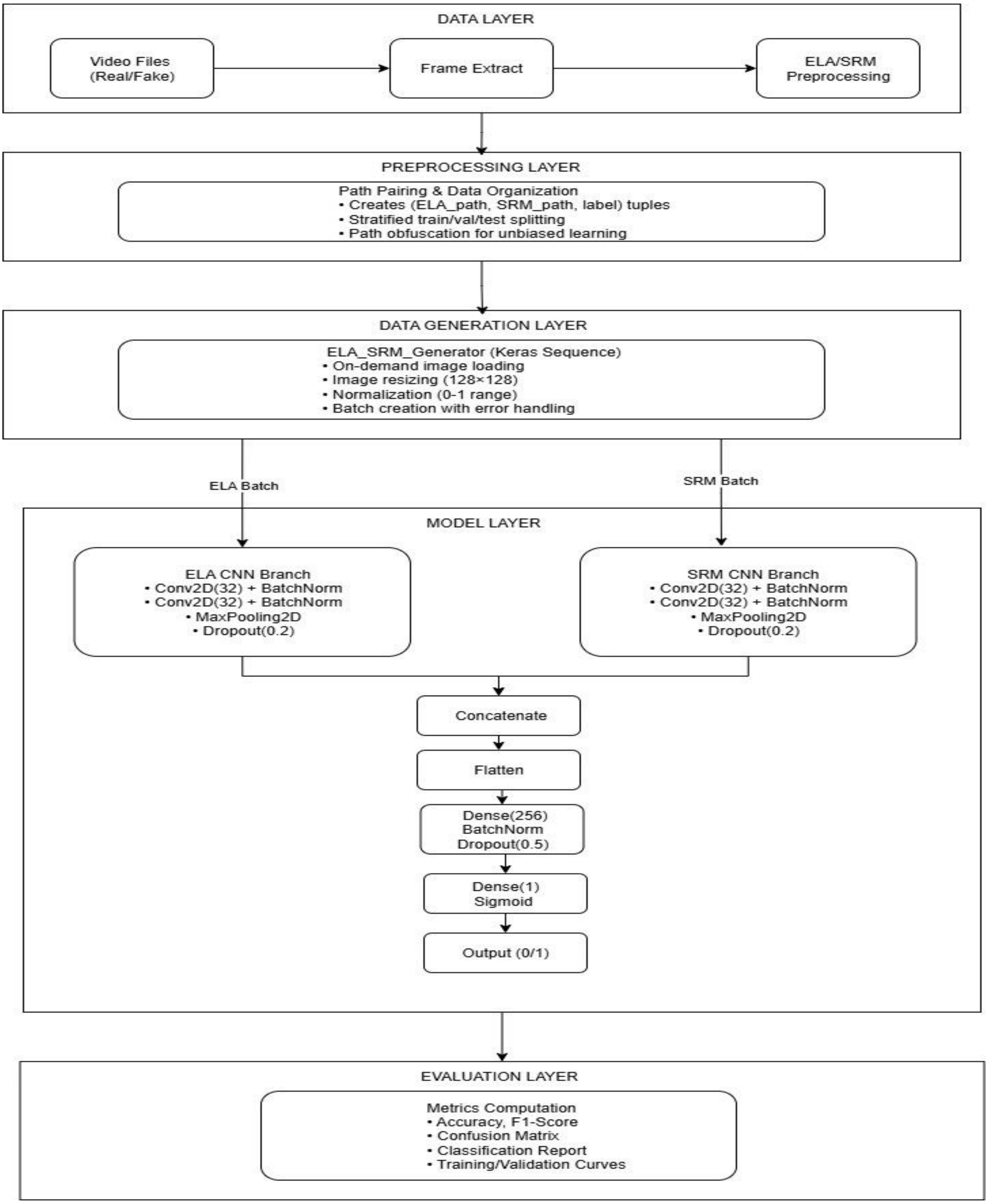
Layer 11: Dropout (0.5)

• Strong regularization

Layer 12: Dense (output)

• 1 unit, Sigmoid

• Produces real vs. fake probability



6. Results

Dataset	Method	Accuracy	AUC
FF-FS	ELA-SRM	98.12	0.9985
FF-FS	DCT-ELA	97.15	0.9958
FF-FS	DCT-SRM	97.77	0.9981
FF-FS	DCT	91.61	0.971
FF-FS	ELA	94.95	0.9875
FF-FS	SRM	99.78	0.9995
FF-F2F	ELA-SRM	98.35	0.9984
FF-F2F	DCT-ELA	96.65	0.9949
FF-F2F	DCT-SRM	98.37	0.9988
FF-F2F	DCT	81.43	0.9007
FF-F2F	ELA	95.57	0.9913
FF-F2F	SRM	98.45	0.9985
FF-NT	ELA-SRM	95.79	0.9918
FF-NT	DCT-ELA	91.15	0.97
FF-NT	DCT-SRM	97.2	0.9957
FF-NT	DCT	73.72	0.8199
FF-NT	ELA	88.94	0.9489
FF-NT	SRM	95.18	0.9882
VidTimit-DFTimit	ELA-SRM	99.99	1
VidTimit-DFTimit	DCT-ELA	99.98	1
VidTimit-DFTimit	DCT-SRM	99.94	1
VidTimit-DFTimit	DCT	91.57	1
VidTimit-DFTimit	ELA	99.99	1
VidTimit-DFTimit	SRM	99.94	1
Celeb-DF	ELA-SRM	99.72	0.9999
Celeb-DF	DCT-ELA	99.89	0.9968
Celeb-DF	DCT-SRM	99.78	0.9988
Celeb-DF	DCT	98.67	0.9987
Celeb-DF	ELA	91.53	0.9707
Celeb-DF	SRM	99.28	0.9994
FF-DF	ELA-SRM	98.33	0.9987
FF-DF	DCT-ELA	97.80	0.9976
FF-DF	DCT-SRM	98.32	0.9988
FF-DF	DCT	93.85	0.9840
FF-DF	ELA	98.89	0.9966
FF-DF	SRM	98.07	0.9979
DFDC	ELA-SRM	99.84	0.9999
DFDC	DCT-ELA	99.66	0.9998
DFDC	DCT-SRM	99.88	0.9999
DFDC	DCT	99.1	0.9992
DFDC	ELA	98.87	0.9992
DFDC	SRM	99.87	0.9998

7. Discussion

The results across all datasets clearly show that **SRM-based models consistently outperform ELA and DCT models**, confirming that frequency-domain noise residuals contain the strongest deepfake fingerprints. Among all single-stream models, **SRM achieves the highest accuracy (up to 99.87%) and AUC (~0.9998)** across datasets like DFDC, VidTimit-DFTimit, and Celeb-DF. ELA performs moderately well but shows drops in more challenging datasets, such as FF-NT and Celeb-DF. DCT alone records the weakest performance across all datasets, especially on FF-NT (73.72% accuracy), indicating its limited standalone discriminative power.

Dual-stream architectures show improved and more stable performance overall. **ELA-SRM fusion consistently ranks among the best**, reaching 98–99.8% accuracy with AUC scores nearing 1.0. This confirms that combining spatial (ELA) and frequency-noise (SRM) features yields complementary benefits. Models combining DCT with ELA or SRM perform better than DCT alone but still underperform compared to ELA-SRM fusion.

Across datasets like FF-FS, FF-F2F, FF-DF, FF-NT, VidTIMIT, Celeb-DF, and DFDC, the **highest accuracy values remain above 98%**, confirming the robustness and generalization of the system across manipulation types (FaceSwap, DeepFakes, NeuralTextures, Face2Face) and real-world datasets (DFDC). Overall, **SRM is the strongest individual feature, and ELA-SRM is the strongest fusion**.

8. Future Scope

1. **Extend to Triple-Stream Fusion (ELA + SRM + DCT):**
   Since DCT improves results when fused, a 3-stream architecture can combine all complementary domains for stronger robustness.
2. **Temporal Modeling for Video-Level Detection:**
   Using 3D-CNN, LSTM, or Optical Flow can capture motion artifacts that frame-level models miss.
3. **Improved Forensic Preprocessing:**
   Introduce additional handcrafted cues such as chroma inconsistencies, color spatial correlation, and noiseprint analysis.
4. **Cross-Dataset Generalization and Adversarial Testing:**
   Evaluate on unseen deepfake methods and adversarially-optimized deepfakes.
5. **Model Optimization for Real-Time Deployment:**
   Pruning, quantization, and ONNX conversion can enable use on edge devices or social media moderation pipelines.

9. Conclusion

Based on the experimental results, the system demonstrates **highly reliable deepfake detection performance across all major datasets**. SRM emerges as the most discriminative single feature, consistently achieving near-perfect metrics. ELA contributes spatial artifact detection, and combining both in a dual-stream model significantly strengthens the classifier. DCT alone is weaker but still adds value when fused with other modalities.

Overall, the **ELA-SRM dual-stream model achieves the best balance**, with accuracy values often exceeding **98–99%** and AUC scores approaching **1.0** across multiple datasets, proving the effectiveness of multi-modal forensic fusion. The pipeline is robust, scalable, and well-suited for real-world deployment with further optimization.

10. References

1. Dey, P., Chakraborty, S., & Chatterjee, K. (2024). *Detection of image tampering using deep learning, error levels and noise residuals*. Neural Processing Letters, 56, 112. https://doi.org/10.1007/s11063-024-11448-9
2. Agarwal, S., Farid, H., Gu, Y., He, M., Nagano, K., & Li, H. (2019). *Protecting world leaders against deep fakes*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW).
3. Yang, X., Li, Y., & Lyu, S. (2019). *Exposing deep fakes using inconsistent head poses*. ICASSP 2019 – IEEE International Conference on Acoustics, Speech and Signal Processing.
4. Han, B., Han, X., Zhang, H., Li, J., & Cao, X. (2021). *Fighting fake news: Two-stream network for deepfake detection via learnable SRM*. IEEE Transactions on Biometrics, Behavior, and Identity Science, 3(3), 320–333.
5. Rossler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., & Nießner, M. (2019). *FaceForensics++: Learning to detect manipulated facial images*. ICCV.
6. Li, Y., Chang, M., & Lyu, S. (2018). *In ictu oculi: Exposing AI-generated fake face videos by detecting eye blinking*. IEEE International Workshop on Information Forensics and Security.
7. Thies, J., Zollhöfer, M., Stamminger, M., Theobalt, C., & Nießner, M. (2019). *Face2Face: Real-time face capture and reenactment of RGB videos*. CVPR.
8. Afchar, D., Nozick, V., Yamagishi, J., & Echizen, I. (2018). *MesoNet: A compact facial video forgery detection network*. IEEE International Workshop on Information Forensics and Security.
9. Fridrich, J., Kodovský, J. (2012). *Rich models for steganalysis of digital images*. IEEE Transactions on Information Forensics and Security.

GitHub Links:

1. https://github.com/souma21122002/Internship2025